

Sign Language Recognition for Patient-Doctor Communication: A Multimedia/Multimodal Dataset

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Abstract - Sign Language translation increases the life quality of deaf people and their social integration. It is an extraordinarily challenging task that requires detecting hand gestures, facial expressions, upper body movements, and finding a temporal relationship among them and the translated words. This paper explores this complex task by a sensor fusion approach, with the objective of defining a medical vocabulary for patient-doctor communication translation for the Italian Sign Language (LIS). Indeed, the availability of a multimodal dataset can be a key factor in supporting machine learning models for Sign Language recognition. In this context, a multimedia/multimodal database has been developed by collecting synchronized data for face, manual, and body from different sensors, i.e. a mm-wave RADAR, a lidar, RGB and RGB-D cameras.

Keywords— Artificial intelligence, Computer Vision, mm-wave RADAR sensor, deep learning, gesture sensors, gesture recognition, sign language, participatory approach, translation.

I. INTRODUCTION

The Italian Sign Language (LIS), recognized by the Italian government in May 2021, is a natural language used by over 40,000 deaf individuals in Italy [1]. Like other sign languages, LIS is a visual-manual language, conveying meaning through body movements and visual perception. Signs involve a combination of manual and body components executed simultaneously. On one hand, manual components, encompassing handshape, palm orientation, location, and movement, may engage one or both hands. On the other hand, body components include facial expressions, mouth actions, torso movement, and eye gaze [2], combining with manual components to collectively form units of meaning (i.e., signs) making LIS a multimodal and multilinear language [3]. A distinctive characteristic of LIS, shared with other sign languages, is its lack of a standardized writing system, making its documentation challenging. The sociocultural aspects, particularly its oral nature and modality, necessitate the capture of LIS through video, resulting in a lack of comprehensive online resources for its documentation.

LIS is not only a minority language, but also an under-documented one, with a scarcity of data collections available [4]. The main aim of this work is to cover this gap in documentation by developing a multimedia/multimodal LIS (MultiMeDaLIS) dataset based on complementary sensor technologies. Specifically, a high-resolution mm-wave RADAR sensor has been used along with a lidar and

RGB/RGB-D cameras to collect synchronized data of a selection of signs.

This work addresses the collection of the MultiMeDaLIS dataset focused on the medical semantic domain to enable translation patientsmedical- staff communications. Specifically, the adopted semantic domain comprises 126 isolated elements, including 100 lexical signs and the 26 signs of the LIS alphabet; overall, the dataset boasts a total of 25,830 sign instances. The selection of signs adhered to identified standards, categorizing signs into two main groups: those related to health and health issues, and those not semantically marked. In fact, while prioritizing signs relevant to health, the dataset also encompasses signs useful for constructing utterances expected in patient-doctor interactions. Ensuring the accuracy and reliability of the MultiMeDaLIS dataset, we selected sign variants from the online dictionary SpreadTheSign. The inclusion of medical terms in the dataset was informed by a review of Italian and international studies on Covid-19-related symptoms and terminology published post-2020 [5].

II. BACKGROUND AND RELATED WORK

In recent years, sign language recognition significantly advanced thanks to the exploitation of different technologies, such as video and RADAR. This Section reviews the recent state of the art of sign language recognition with reference to the adopted technologies and the available datasets.

A. Technologies for Sign Language Recognition

Automatic sign language recognition approaches can be broadly categorized into two main classes: those relying on specialized hardware and those utilizing visual information. The former employs specialized hardware, such as gloves equipped with accelerometers to capture the precise movements of the hands. While these systems can provide detailed data, they are often considered intrusive and can hinder the natural flow of communication. Moreover, they are unable to capture the full spectrum of sign language, which includes full-body movements and facial expressions.

In contrast, vision-based approaches leverage visual information captured by cameras, including RGB, depth, infrared, or a combination thereof. These methods are less obtrusive for users, as they do not require wearing any special equipment. Common approaches combine visual sensors with statistical methods (e.g., [6] use Hidden Markov Models for continuous sign recognition) or, more recently, deep learning

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models, which have proved to be particularly effective at processing visual signals. For instance, recurrent neural networks [7] and convolutional neural networks (CNNs) [8] have been employed to perform word- and sentence-level translation. Convolutional model also provides a natural way to process multi-modal visual data, e.g., by combining RGB and depth [9], and to integrate both manual and non-manual components through the fusion of multi-cue information via CNNs [10], [11].

However, it has been noted that RGB-based sign language recognition systems may raise privacy concerns, especially in scenarios where visual data must be processed in the cloud or used for training/fine-tuning machine learning models, thus limiting their applicability in public environments [12]. At the same time, radio frequency (RF) sensors emerged as a promising alternative, offering privacy protection while enabling American sign language (ASL) recognition through innovative data representations, such as 2D/3D RF data and optical flow analysis [12], [13]. Most approaches in the literature apply deep learning approaches to a variety of RF modalities: ultrawideband (UWB) [14], [15], doppler [16], [17], continuous-wave (CW) [18], [19], micro-Doppler [20], [21], [22], frequency modulated continuous wave (FMCW) [23], [24], [25], [26], multi-antenna systems [27], millimeter wave [28], [29], [30], [31], [32].

Currently, RADAR-based solutions have demonstrated robust performance across different environmental conditions, showing the importance of including this sensor technology in data collection efforts.

B. Sign Language Datasets

Due to the success of applying deep learning techniques to sign language recognition, the availability of diverse and comprehensive datasets to develop and evaluate models effectively becomes a critical factor. Datasets based on video capture have been central in this field, supplying extensive data for training and testing models.

The Word-Level American Sign Language (WLASL) dataset [33] is a notable contribution, containing videos of 2,000 ASL words performed by over 100 signers. The MS-ASL dataset [34] provides 25,000 video clips of 1,000 signs, making it one of the largest ASL datasets available. The LSA64 dataset [35] offers a specialized set of 3,200 videos for 64 Argentine Sign Language signs. The SIGNUM dataset [36] provides a comprehensive German Sign Language corpus with 450 signs and 780 sentences across more than 30,000 videos.

The AUTSL dataset [37] stands out for its multi-modal nature, containing over 38,000 video samples of 226 isolated signs in Turkish Sign Language, complemented by RGB, depth, and skeletal data from Kinect. Similarly, the DGS Kinect 40 dataset [38] includes 3,186 videos of 40 German Sign Language signs, with additional skeletal data captured by a Kinect sensor. The RWTH-PHOENIX-Weather 2014 dataset [39] contributes over 75,000 videos of continuous sign language with hand pose annotations, supporting both discrete and continuous sign language recognition.

In comparison, RADAR-based datasets for sign language are less extensive, often with fewer signs and total number of samples, which significantly limits the possibility of deep model training. For instance, most RADAR datasets include between 4-5 signs [40], [41] and 10-20 signs [42], [43], [44],

[45]. In terms of samples per dataset, the ones with fewer signs feature significantly more repetitions (220 for a single subject in [45] and 250 for five subjects in [40]) than datasets with a larger variety of signs (up to 10 repetitions for five subjects in [44]), making them even less effective for model training.

In this work, we aim to fulfil the need for an expansive multi-modal dataset that merges the advantages of camera and RADAR data, matching the size of camera-based datasets and incorporate the additional features and robustness of RADAR data, supporting the creation of more precise and widely applicable sign language recognition systems that function effectively under various conditions.

III. DATA ACQUISITION AND METHODOLOGY

A. Selection of Signs

As already mentioned in Section II, a decision was made to semantically limit the signs included in the dataset to the medical domain. The recorded signs are consequently divided into macro categories: semantically marked and unmarked signs. All signs were recorded in isolation. In identifying the signs to include in the dataset, it was decided to consider other existing corpora resources for LIS [48], [49], [50], [51] and spoken Italian [52], [53], as well as frequency vocabularies for spoken Italian [19].

To our knowledge, no digital collections of LIS with a specific focus on the medical domain are available. Consequently, in the absence of collections of this kind, we sourced semantically marked terms for the MultiMeDaLIS dataset through a review of common COVID-19 symptoms in both Italian and English [54], [55]. Additionally, terms related to both diagnosis and treatment solutions are considered [53]. To deepen our understanding of the experience of Italian deaf people in medical contexts, a set of interviews was conducted respectively with a doctor, a deaf person, and a sign language interpreter.

For the sake of completeness, Table I summarizes labels of the selected signs, grouped in various categories.

B. Selection of Tools for Data Collection

Previous experiences in data collection of sign-language data highlight how crucial are both the identification of tools for capturing the MultiMeDaLIS dataset and a well-defined methodological approach. The visual-manual modality of LIS accentuated the importance of capturing both manual and body components and their positions in the 3D space.

The adopted methodological approach has been grounded in the following key principles to select the recording tools:

- *portability*: the designed set-up should facilitate data acquisition across diverse settings, enabling signers to easily express themselves in real-world contexts;
- *non-invasiveness*: the data acquisition process should not interfere with the signer's natural movements or comfort, ensuring that the collected data reflects authentic sign language usage;
- *multimodality*: different sensor technologies provide specific advantages and shortcomings; hence it is critical to integrate multiple acquisition modalities to fully represent the complexity of sign language communication;

TABLE I. LIST OF THE SIGN ENGLISH LABELS GROUPED IN CATEGORIES

Categories	Sign English Labels		
Body parts	Arm	Back	Chest
	Ear	Eye	Feet
	Forehead	Hand	Head
	Heart	Mouth	Muscle
	Neck	Shoulder	Throat
Medical actions / Conditions	Anxiety	Bad	Cough
	Deaf	Difficulty	Illness
	Infection	Injection	Medicine
	Nausea	Pain	Pale
	Positive	Stomach	Sweat
	Symptoms	Tiredness/ Fatigue	Vertigo
	Vomit		
Medical facilities / Professionals	Doctor	Emergency room	Hospital
	Pharmacy		
Time expressions	Day	Morning	Night
	Noon	Time	Tomorrow
	Yesterday		
Health-related terms	To ask	To breathe	To buy
	To call	To communicate	To cure
	To disinfect	To eat	To have a cold
	To have to	To smoke	To stay
Pronouns	Me	You	
Family terms	Father	Mother	Son/ Daughter
Interrogatives	How	When	Where
	Why		
Negation	No		
Quantifiers	A lot		
Conjunction	If		
Affirmative response	Yes		
Instruments/ tools	Mask	Syrup	Thermometer
	Vaccine		
Other	Ambulance	Emergency room	Groin
	Heat	Helicopter	House
	Light	Name	Noise
	Person	Problem	Shoulders

- *replicability*: as explained above, the protocol for data collection should be suitable to be employed by users in their everyday life, without introducing data distribution shifts;

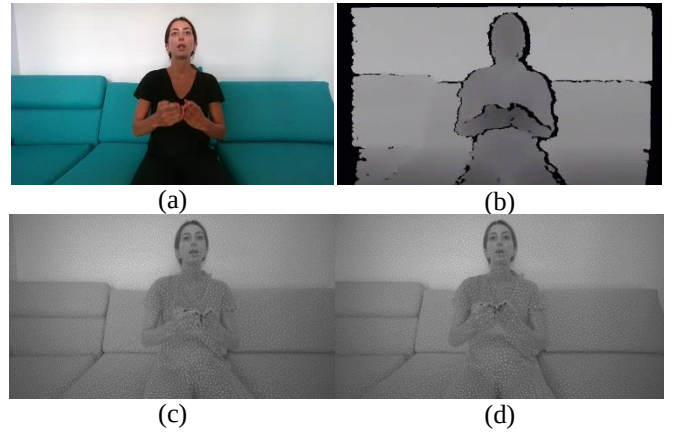


Fig. 1. Data acquisition: (a) RGB, (b) depth map; (c) infrared stereoscopic with laser projection (left vision), (d) infrared stereoscopic with laser projection (right vision)

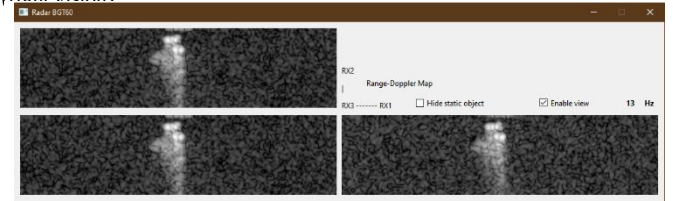


Fig. 2. Range doppler map from the three radar antennas.

- *high-quality data*: the recording equipment must be

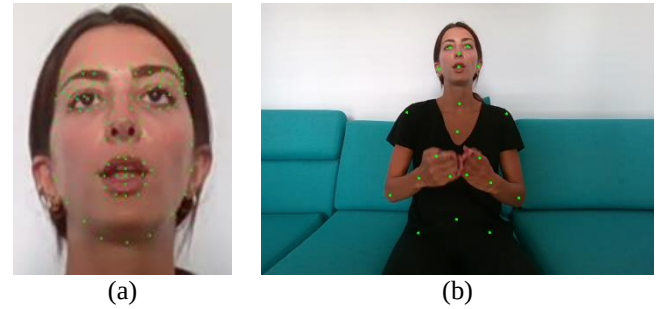


Fig. 3. Face (a) and body (b) tracking output.

capable of capturing data with an elevated level of detail and clarity, to facilitate accurate analysis and interpretation of the sign language.

The dataset has been built by collecting synchronized data recorded by means of the following tools:

- 1) Infineon XENSIV 60-GHz RADAR [57];
- 2) Intel Realsense D455 depth camera [58];
- 3) Microsoft Kinect v1 motion camera [59];
- 4) Stereolabs Zed 2 stereo camera **Error. L'origine riferimento non è stata trovata.**

RADAR technology enables capturing the motion and position of the signer's hands and fingers (i.e., manual data), while ensuring users privacy since the visual image is not recorded. RADAR is also able to capture subtle movements, works in absence of physical contact and is not affected by ambient light, allowing for flexibility of recording environments. For these reasons, the use of high-resolution mm-wave radar gives the dataset a high degree of novelty. Realsense D455 depth camera is used to record several types of body data, while facial expressions are captured by means of Kinect v1. Finally, Zed 2 camera ensure high-resolution data

capture, comprehensively encompassing both manual and body components.

For the sake of completeness, Table II details each adopted recording tool and the corresponding captured data.

Fig. 1 shows the acquired data with different technologies, while the range doppler map from the three radar antennas is depicted in Fig. 2. The body and face tracking output are reported in Fig. 3.

C. The Recording Protocol

To ensure the consistency and reliability of the collected data a recording protocol was developed. A total of 200 recording sessions were executed to strengthen the robustness of the dataset. Additionally, five extra sessions were conducted to account for any recordings that were unsuccessful or inadequate, thereby ensuring a complete set of 200 high-quality recordings. Although the protocol is designed to be reproducible by multiple individuals, it has been initially validated and subjected to preliminary analysis using AI models with a single non-native LIF signer who possesses extensive experience in Italian Sign Language.

To facilitate data recording during data collection, a customized software application was designed. This software was uploaded to a desktop computer and a keyboard was repurposed into a two-pedal control device. The pedal board was equipped with two pedals, one for forward navigation (right pedal) and one for backward navigation (left pedal) within the data collection software. This arrangement allowed users to control the data recording process by using their feet, minimizing upper body movement, and reducing the potential for altering the neutral recording position.

During data collection sessions, the signer is presented with one of the 126 Italian labels or letters of the Italian alphabet on the computer screen. Each sign recording begins with the participant in a neutral posture, and they must return to this position after completing each sign. The signer uses the right pedal to indicate the completion of the sign. However, if the participant makes an error (e.g., produces an unclear sign or presses the right pedal prematurely), they can use the left pedal to restart the recording for that specific sign, allowing for re-recording as many times as needed.

The software employs background color variation to differentiate the preparation and recording phases. A yellow background marks the preparation phase, giving participants time to position themselves for the upcoming sign, while a green background marks the recording phase, prompting the participant to begin signing, as shown in Fig. 4.

D. Discussion on Usability and Reliability

We assess the reliability of the proposed dataset through validation sessions with ten native LIS signatories from various regions of Italy, from north to south. This group includes signers of different ages, genders and regional backgrounds. Participants ranged in age from 18 to 54: 40% were 35 to 44, 30% were 25 to 34, 20% were 45 to 54, and 10% were 18 to 24. The group was made up of 6 men and 4 women, all deaf signers, coming from different regions of Italy such as Lombardy, Veneto, Tuscany, Marche, Lazio, Campania, Sicily and two from Puglia, with one participant from Tunisia. Overall, signers provided a positive feedback, confirming the accuracy and cultural relevance of the signs collected, ensuring that the dataset is technically valid and

TABLE II. LIST OF RECORDING TOOLS AND RELATED CAPTURED DATA

Recording Tool	Captured Data
Infineon XENSIV 60-GHz RADAR	Various types of manual data at 13 fps, including: ▪ Time-Domain data. ▪ Frequency-Domain data. ▪ Range Doppler Map data. ▪ Range Doppler Map data with Moving Target Indication (MTI) collected at the same frame rate.
Intel Realsense D455 Depth Camera	Various type of body data, including: ▪ n.2 infrared (IR) pictures in stereoscopic mode at 848x480 pixels, 30 fps. ▪ n.1 RGB picture at 1280x720 pixels, 30 fps. ▪ n.1 a 16-bit depth map at 848x480 pixels, 30 fps. ▪ n.1 filtered depth map at 424x240 pixels, 30 fps. ▪ n.1 filtered depth map aligned with the RGB image at the same resolution and frame rate. ▪ n.1 3D point cloud at 848x480 pixels and 30 fps. ▪ n. 68 face tracking points at 30 fps.
Microsoft Kinect v1 Motion Camera	Various data types specifically related to facial expression: ▪ n.1 RGB image at 640x480 pixels, 30 fps. ▪ n.1 16-bit depth map at 640x480 pixels 30 fps ▪ 87 2D face tracking points at 30 fps.
Stereolabs Zed 2 Stereo Camera	Employed to capture the following data types: ▪ n.2 RGB images in stereoscopic mode at 1920x1080 pixels, 25 fps. ▪ n.2 32-bit depth maps in stereoscopic mode at 1920x1080 pixels, 25 fps. ▪ n.2 3D point clouds in stereoscopic mode at 1920x1080 pixels, 25 fps. ▪ 18 3D body tracking points at 25 fps.

socially and linguistically appropriate. In order to favour usability, in addition to carrying out the next phase of expanding the dataset and consequent publication of a new



Fig. 4. Background color variation to differentiate the preparation (yellow) and recording (green) phases of the sign.

artificial intelligence method, the complete dataset will also be released, sharing with the scientific community a valuable resource for further research and development in the field.

IV. CONCLUSIONS

This paper has presented the collection of a LIS database focuses on medical vocabulary of 100 words for patient-

doctor communication, as well as 26 letters of the alphabet; for each sign, 205 repetitions have been performed from one LIS signer. As a result, for the first time, more than 25,000 sign instances have been included in a multimodal dataset, recording data from face, hands and body through a sensor fusion approach collecting synchronized data from different complementary sensors, i.e. a mm-wave RADAR, a lidar, RGB/RGB-D and stereoscopic cameras. Preliminary analyses have been conducted on the collected data using state-of-the-art machine learning and deep learning methods for each type of acquisition, including RGB, depth, and body tracking joints data. These analyses have yielded promising accuracy levels, serving as a baseline for future work. These results provide a solid foundation for developing an advanced AI model that integrates all available data types, enhancing the overall performance and accuracy of sign language recognition and translation. The next phase of this research involves replicating the protocol with an additional ten native LIS signers to capture diverse signing styles and nuances, thus enhancing the generalizability of the dataset. Following this, advanced AI tools will be employed to develop a comprehensive sign language recognition and translation system in medical settings. The objective is to create a model that primarily utilizes radar data, while supplementing it with depth and body tracking joints data during the training phase to improve generalization. This approach seeks to enhance recognition accuracy and ensure robust performance by leveraging the strengths of each data modality. Furthermore, it addresses challenges related to hospital data, including consent issues, privacy concerns regarding facial recognition, and the risks associated with non-localized data processing, such as data leaving the hospital premises and potential delays in data deletion.

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